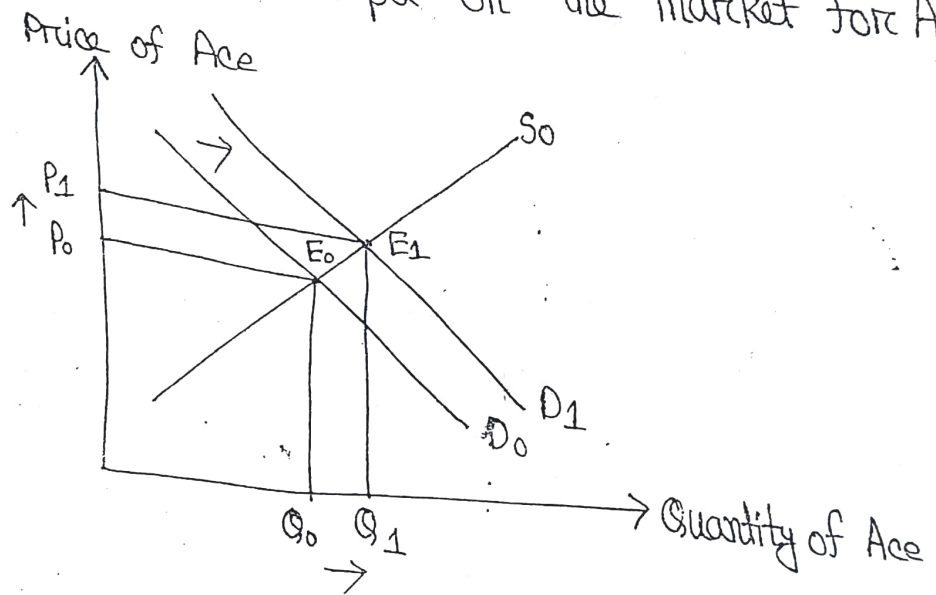


Using a supply and demand graph and assuming perfect competition, show and explain the effects on equilibrium price and quantity of the following :

(a) Increased price of Napa on the market for Ace.

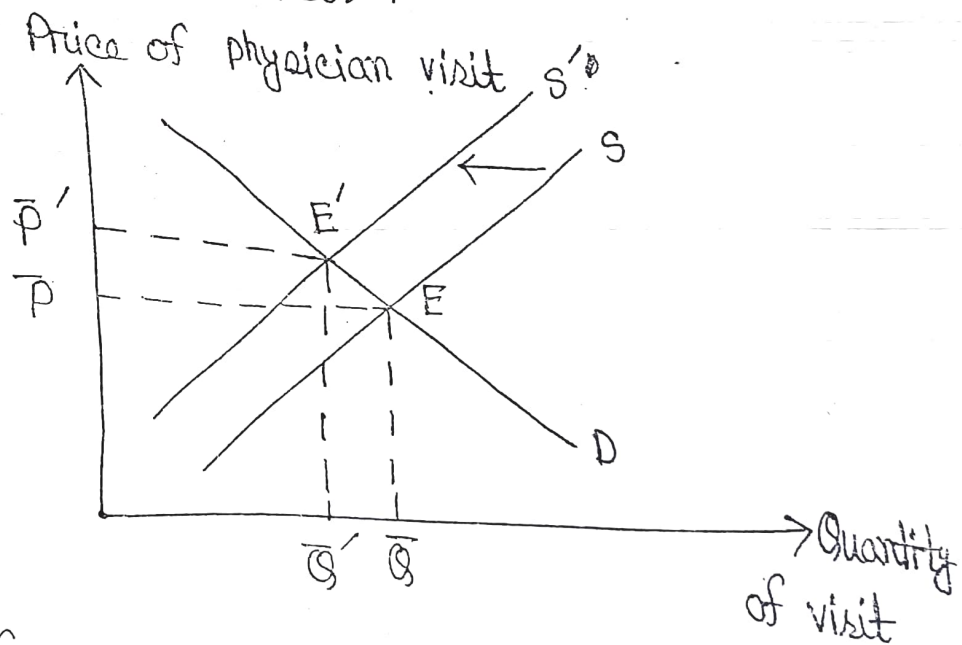
Substitute



Suppose, initially the market for Ace ~~is~~ was in equilibrium at point E_0 where Ace demand curve D_0 and Ace supply curve S_0 ~~are~~ were intersect and price of Ace was P_0 and quantity supplied of Ace was Q_0 . Now price of Napa has increased and Ace is a substitute good of Napa. So, as the price

of supply increases so that demand for Ace will be increased which shifts the Ace demand curve rightward from D_0 to D_1 . As a result ^{equilibrium} shifts from E_0 to E_1 and price of Ace will increase from P_0 to P_1 and quantity supplied of Ace will increase from Q_0 to Q_1 .

☒ Increased cost of medical education on the market for physician services.

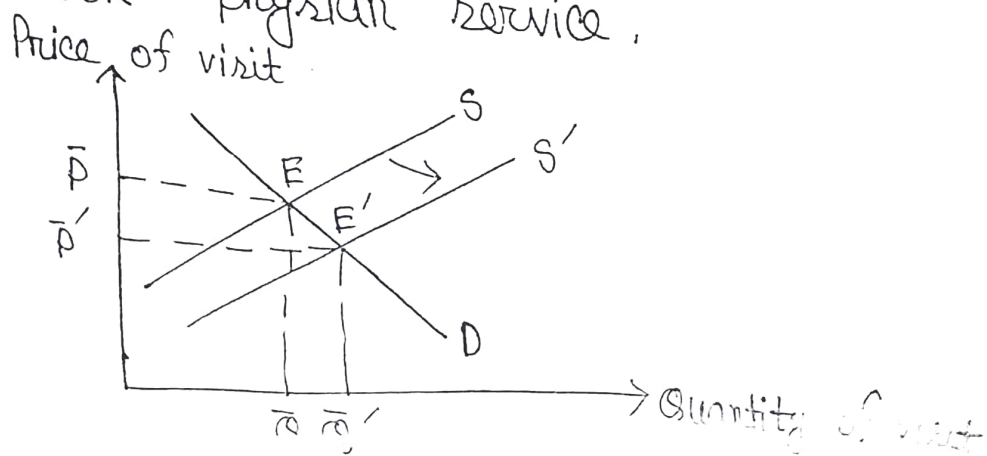


~~Increased cost of medical education~~
 Suppose, initially equilibrium established at point E on the market for physician services where demand

and supply curve intersect. The equilibrium physician visit was $\bar{P}$ and quantity of visits was $\bar{Q}$.

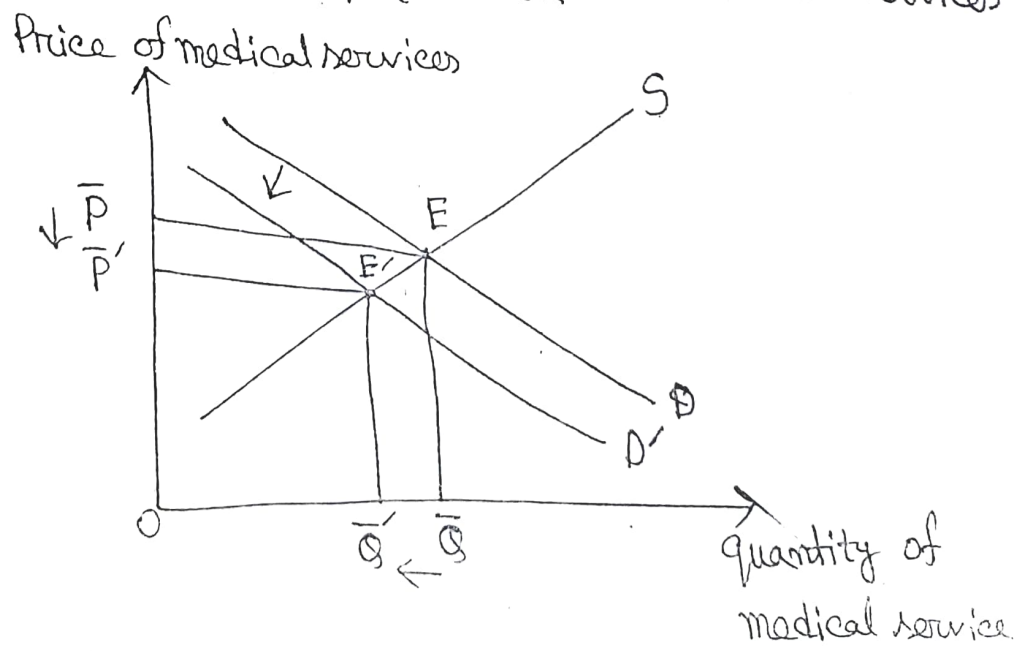
Now, an increased cost of medical education will shift the supply curve ^{leftward} from S to S' . As a result equilibrium shifts from E to E' , and price of physician visit increases from $\bar{P}$ to $\bar{P}'$. As price of physician visit increases so that the quantity of visit decreases from $\bar{Q}$ to $\bar{Q}'$.

② Increased graduations of new doctors on the market for physician service.



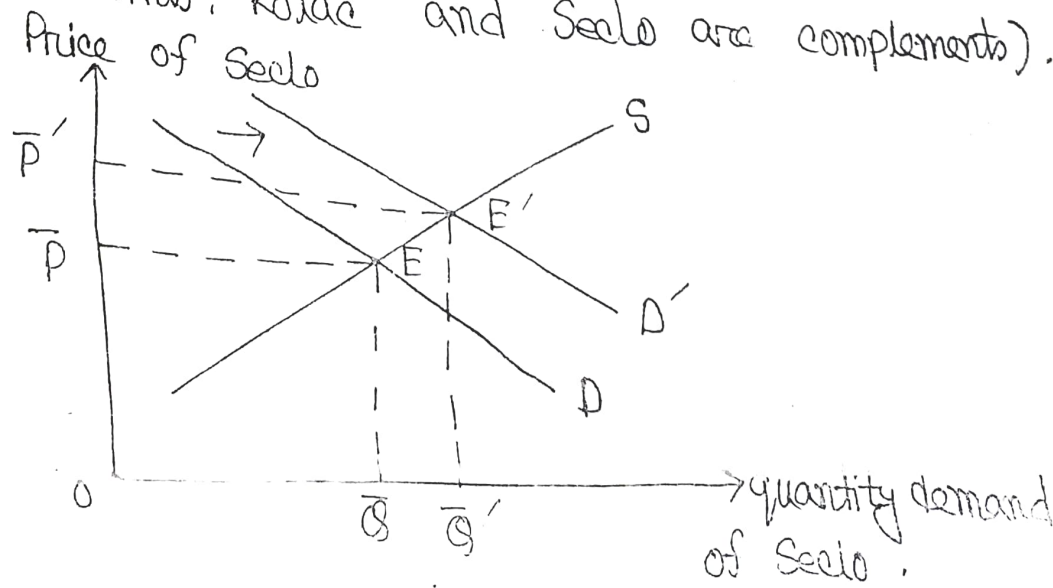
Suppose increased graduations of new doctors on the market for physician services will shift the supply curve to the right from S to S' , as a result equilibrium also shifts from E to E' . Now equilibrium price of visit decreases from $\bar{P}$ to $\bar{P}'$. As price of visit decreases so that the quantity of visit increases from $\bar{Q}$ to $\bar{Q}'$.

④ the virtual elimination of smoking in the population on the market for medical services



the virtual elimination of smoking will improve people's health status so that people now going less to doctors. As a result the demand for medical services reduces which shift the demand curve leftward from D to D' and, as a result equilibrium shifts from E to E' . As equilibrium quantity of medical service reduces from $\bar{Q}$ to $\bar{Q}'$, the price of medical services will be decreased from $\bar{p}$ to p' .

(e) Decreased price of Rolac ~~on the~~ on the market for Seelo (Hint: Rolac and Seelo are complements).



Ans
As Rolac and Seelo are complements,
decrease in the price of Rolac will
increase the demand of Seelo so that
the demand curve ~~for~~ on the market for
Seelo will shift to the right from
 D to D' and as a result equilibrium
also shifts from E to E' , and price
of Seelo increases from $\bar{P}$ to $\bar{P}'$ and
quantity demanded of Seelo increases from
 $\bar{Q}$ to $\bar{Q}'$.